

Solution to the Daily Limit

Based on the problem provided in the image dailyintegral-limit-limit-1x1-2026-08-13.jpg, we are tasked with evaluating a limit involving a sequence $(a_n)_{n \geq 1}$ defined recursively as:

$$a_{n+1} = \frac{2a_n}{4a_n + 3}, \quad n \geq 1, \quad a_1 = 1$$

$$\text{Evaluate: } \lim_{n \rightarrow \infty} \left(\frac{3}{2}\right)^n a_n$$

1. Transforming the Sequence

To simplify the non-linear recurrence relation, we can take the reciprocal of both sides. First, we note that $a_1 = 1 > 0$, and if $a_n > 0$, then $a_{n+1} > 0$. By induction, $a_n > 0$ for all $n \geq 1$, ensuring taking the reciprocal is valid.

$$\frac{1}{a_{n+1}} = \frac{4a_n + 3}{2a_n} = 2 + \frac{3}{2a_n}$$

Let us define a new sequence (b_n) such that $b_n = \frac{1}{a_n}$. The recurrence relation now becomes a linear sequence:

$$b_{n+1} = \frac{3}{2}b_n + 2, \quad b_1 = \frac{1}{a_1} = 1$$

2. Solving the Linear Recurrence

To solve this linear recurrence relation, we find the fixed point L by setting $L = \frac{3}{2}L + 2$:

$$L - \frac{3}{2}L = 2 \implies -\frac{1}{2}L = 2 \implies L = -4$$

We define an auxiliary sequence $c_n = b_n - L = b_n + 4$. We can now determine its recurrence relation:

$$c_{n+1} = b_{n+1} + 4 = \left(\frac{3}{2}b_n + 2\right) + 4 = \frac{3}{2}b_n + 6 = \frac{3}{2}(b_n + 4) = \frac{3}{2}c_n$$

This demonstrates that (c_n) is a geometric sequence with a common ratio of $r = \frac{3}{2}$ and a first term $c_1 = b_1 + 4 = 1 + 4 = 5$. The general term for c_n is therefore:

$$c_n = 5 \left(\frac{3}{2}\right)^{n-1}$$

Substituting this back to find b_n and subsequently a_n :

$$b_n = c_n - 4 = 5 \left(\frac{3}{2}\right)^{n-1} - 4$$

$$a_n = \frac{1}{b_n} = \frac{1}{5 \left(\frac{3}{2}\right)^{n-1} - 4}$$

3. Evaluating the Limit

Now we substitute our explicit formula for a_n into the limit we wish to evaluate:

$$\lim_{n \rightarrow \infty} \left(\frac{3}{2}\right)^n a_n = \lim_{n \rightarrow \infty} \frac{\left(\frac{3}{2}\right)^n}{5 \left(\frac{3}{2}\right)^{n-1} - 4}$$

To evaluate this limit, we can divide the numerator and the denominator by $\left(\frac{3}{2}\right)^n$:

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{5 \left(\frac{3}{2}\right)^{-1} - 4 \left(\frac{2}{3}\right)^n} &= \lim_{n \rightarrow \infty} \frac{1}{5 \left(\frac{2}{3}\right) - 4 \left(\frac{2}{3}\right)^n} \\ &= \lim_{n \rightarrow \infty} \frac{1}{\frac{10}{3} - 4 \left(\frac{2}{3}\right)^n} \end{aligned}$$

Since $\left|\frac{2}{3}\right| < 1$, the term $\left(\frac{2}{3}\right)^n$ approaches 0 as $n \rightarrow \infty$. Thus:

$$\lim_{n \rightarrow \infty} \left(\frac{3}{2}\right)^n a_n = \frac{1}{\frac{10}{3} - 0} = \frac{3}{10}$$

Conclusion

The final limit of the sequence is:

$$\lim_{n \rightarrow \infty} \left(\frac{3}{2}\right)^n a_n = \frac{3}{10}$$